Weekly Report 6

ML Code Crafters

Project 7: Offline Track association problem [UAV videos]

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Work Done This Week
Status: Completed initial implementation and started testing phase

1. The Goal for This Week

This week we needed to have a completion of what we were planning to do. Our attention was on the coding of the central offline association pipeline and running it on test sample outputs of UAV tracking. We wanted to test the workability of our two-stage association strategy.

2. Plan for Implementation

We were able to get started on the Two-Stage Association logic.

Stage A: Construction of Similarity Matrix

We did the similarity computation between lost and new tracklets by the use of the three scoring layers:

1. **Motion Score (Kalman Prediction):**
   In order to make a prediction about the future whereabouts of lost tracklets, we used a simple Kalman prediction model. Those scores of motion were assigned by means of distance-based thresholding.

2. **Visual Score (Cosine Similarity):**
   To obtain the feature vectors we used a pre-trained model and calculated the similarity between tracklets by using co-sin similarity to identify visual consistency.

3. **IOU Score (Spatial Overlap):**
   The IOU of the overlapping bounding boxes was used to guarantee the spatial continuity of the detections.

A weighted similarity score was then created with all three scores as the weighted score on a tracklet pair.

Stage B: Solving the Assignment

The Hungarian Algorithm, implemented in SciPy, helped us to perform optimal matching of new tracklets and lost ones. This guaranteed optimum assignments throughout the world rather than greedy matching.

3. Techniques & Tools We Are Using

- **Python (Core Logic)**: Wrote the entire pipeline in NumPy and SciPy.
- **Deep Learning (Re-ID):** An existing model was utilized to get feature embeddings.
- **OpenCV:** It is applied in dealing with bounding boxes and initial motion adjustments.
- **SciPy Linear Assignment:** Tracklet tracklet optimal matching.

4. What We Expect to Learn (Our "Hypothesis")

- The motion score was doing well when tracking short-term gaps.
- Tracklets separated by time were re-linked with the help of visual similarity.
- The accuracy of matching was improved when all three scores were used compared to that of using a single measure.

5. Challenges We Anticipate

- **Parameters Tuning:** It was not easy to choose the correct weights of Motion, Appearance, and IOU, and this involved a number of trials.
- **False Matches:** Sometimes, false matches were made between similar looking vehicles in heavy traffic.
- **Computation Time:** Construction of similarity matrix of all tracklets was time-consuming, particularly with long videos.

6. Planning for Next Week

- **Optimize Performance**: time window filtering can be used to minimize the comparisons made.
- **Get More Accurate:** Fine Tune Scoring Weights and Thresholds with a Better Match.
- **Evaluation Metrics:** The IDF1 and MOTA scores measure the improvements.
- **Multi-Video Testing:** This is an additional test that is done to verify the strength of the pipeline to make sure that the test is run on a variety of UAV clips.